

TRECYA DEWI KUSUMANINGRUM

Frontend & Software Engineer | Informatics Graduate

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SUMMARY

Informatics graduate from Universitas Al Azhar Indonesia (awaiting graduation ceremony), with internship experience at PT Jalin Pembayaran Nusantara building internal web tools for the engineering team. Hands-on with React, Next.js, TypeScript and Tailwind CSS across internship, freelance and personal projects, from full-stack web apps to PWAs and desktop widgets. Background in deep learning research for Indonesian speech recognition, with a strong interest in AI engineering and AI-assisted development.

EXPERIENCE

Technical Writer Intern - PT Jalin Pembayaran Nusantara

2026 - Present

Switching & Innovation Solution Team

- Built a **React-based KPI tracking web app** for the development team with **role-based access for 5 roles**, translating functional and technical specification documents (FSD/TDD) into working features.
- Designed scoring logic driven by **JIRA activity data**; identified inconsistencies between specification documents and escalated them to the Product Owner for resolution.
- Developing an **aging/SLA monitoring dashboard** from JIRA tickets, tracking turnaround time per person, unit and institution.
- Reviewed technical documents and contributed to technical presentation materials for cross-team projects.

Part-time Barista - Marila, El Primero, December & Arctic Coffee

Dec 2020 - Oct 2023

Jakarta Selatan, BSD & Gading Serpong

- Worked in fast-paced, small-team environments across four cafés while studying; handled cashier and payments accurately and kept service quality consistent during peak hours.

PROJECTS

NR Order System - Freelance, Full-Stack

Next.js · TypeScript · Node.js · SQL

- Built a web-based ordering system for an online gaming community end to end: responsive UI with Tailwind CSS, Node.js API and SQL database.

Pink Panther GTA V Marketplace - Personal, Full-Stack

Next.js · Node.js · TiDB · Vercel

- Built and deployed a marketplace web app on Vercel backed by TiDB Cloud; debugged and resolved CORS and SSL configuration issues in production.

Bipo - Desktop Widget

Tauri · JavaScript

- Frameless, always-on-top pixel-art robot widget whose expressions react to live weather data and laptop battery status.

Milo - Progressive Web App

HTML · CSS · JavaScript

- Installable Tamagotchi-style pixel cat game with custom-drawn sprites, refined through many design iterations.

Interactive Browser Mini Games - Freelance

HTML5 Canvas · Vanilla JS

- Built a maze game from scratch without a game engine: character movement, collision detection, background music and staged pop-up messages.

RESEARCH

Undergraduate Thesis - BiGRU vs BiLSTM in CRNN-CTC for Indonesian Speech

2026 · Python · PyTorch

Recognition

- Compared two bidirectional recurrent layers in a Deep Speech (CRNN-CTC) architecture, trained on **30,256 utterances** from Mozilla Common Voice Indonesian.
- Best result: **BiLSTM reached 26.36% WER and 6.59% CER**, outperforming BiGRU (28.13% WER, 7.01% CER). Submitted as a scientific journal article.

SpamGuard-ID - Group Project

IndoBERT · NLP

- Indonesian spam classification using IndoBERT, written up in a journal article format.

EDUCATION

Bachelor of Computer Science (S1 Informatika) - Universitas Al Azhar Indonesia

2026 · Awaiting graduation

SKILLS

Frontend: React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, HTML5 Canvas, PWA

Backend & Tools: Node.js, SQL, Firebase, Tauri, Git/GitHub, Vercel, JIRA

Machine Learning: Python, PyTorch, deep learning for speech recognition, NLP (IndoBERT)

Languages: Indonesian (native), English (professional working proficiency)

ORGANIZATIONS

HMIF (Informatics Student Association) - Communication Dept. • **BEM FST** - Student Development Dept.; Treasurer, Dekan Cup FST • **Saman Dance Club** - active member, multiple competition wins